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GATE 2023: Official ECE Paper with Step-by-Step Solutions

Complete Question-by-Question Analysis & Detailed Mathematical Derivations

Exam Session: February 2023 • **Paper Code:** EC • **Total Marks:** 100 Marks

Q.1 (1 Mark - MCQ) • Digital Circuits - Logic Simplification

Official GATE Examination Question & Solution
The minimal sum-of-products expression for the Boolean function F(A, B, C) = A'B + BC' + BC + AB'C is:
(A) B + AC (B) A'B + C (C) B + A'C (D) AB + C

Correct Option: (A) B + AC

Comprehensive Step-by-Step Solution:
1. Combine terms 2 and 3: BC' + BC = B(C' + C) = B.
2. Expression becomes: A'B + B + AB'C.
3. Absorption law: A'B + B = B (since B absorbs A'B).
4. Now we have: B + AB'C.
5. By rule of distribution: B + AB'C = (B + AB') * (B + C) = (B + A) * (B + C) = B + AC.
6. Result is **B + AC**.

Q.2 (1 Mark - NAT) • Control Systems - Steady State Error

Official GATE Examination Question & Solution
A unity feedback control system has open-loop transfer function G(s) = 50 / [s (s + 5)]. The steady-state error for a unit ramp input r(t) = t u(t) is _____.

Correct Answer: 0.1

Comprehensive Step-by-Step Solution:
1. System is Type 1 because of the single pole at the origin s = 0.
2. Velocity error constant K_v = lim_{s->0} s * G(s) = lim_{s->0} s * [50 / (s(s+5))] = 50 / 5 = 10.
3. Steady-state error for unit ramp: e_ss = 1 / K_v = 1 / 10 = **0.1**.

Q.3 (2 Marks - MCQ) • Network Theory - Thevenin's Equivalent

Official GATE Examination Question & Solution
A 10 V DC voltage source with internal resistance 4 Ohms is connected to a circuit with an independent current source of 2 A in parallel with a 6 Ohm resistor. The Thevenin equivalent resistance R_th seen by the load connected across the output terminals is:
(A) 10 Ohms (B) 2.4 Ohms (C) 4.8 Ohms (D) 1.5 Ohms

Correct Option: (B) 2.4 Ohms

Comprehensive Step-by-Step Solution:
1. To find R_th, deactivate all independent sources:
- Replace 10 V voltage source with a SHORT circuit.
- Replace 2 A current source with an OPEN circuit.
2. Looking into the terminals, the 4 Ohm resistor and 6 Ohm resistor are connected in PARALLEL.
3. R_th = (4 * 6) / (4 + 6) = 24 / 10 = **2.4 Ohms**.

Q.4 (2 Marks - NAT) • Communication Systems - Channel Capacity

Official GATE Examination Question & Solution
An analog telephone channel has bandwidth B = 3.4 kHz and SNR = 30 dB. According to Shannon's capacity formula, the maximum data rate in kbps (rounded to one decimal place) is _____.

Correct Answer: 33.9 kbps

Comprehensive Step-by-Step Solution:
1. Convert SNR from dB to linear: SNR = 10^(30/10) = 10^3 = 1000.
2. Shannon capacity: C = B * log_2(1 + SNR) = 3400 * log_2(1 + 1000) = 3400 * log_2(1001).
3. log_2(1001) = ln(1001) / ln(2) = 6.908755 / 0.693147 ≈ 9.9672.
4. C = 3400 * 9.9672 = 33,888 bps = **33.9 kbps**.